

Abstract

The Smart Curight system represents a significant advancement in emergency response technology, offering a comprehensive multispecialty solution for tracking and managing emergency situations. Designed to integrate seamlessly with existing emergency infrastructure, Smart Curight leverages state-of-the-art tracking technologies, real-time data analytics, and advanced communication tools to enhance the efficiency and effectiveness of emergency response teams. The system provides a unified platform for managing various types of emergencies, including medical, fire, and natural disasters, enabling rapid coordination and response. Key features include GPS-based tracking of emergency vehicles, real-time incident monitoring, and automated alert systems. By facilitating improved situational awareness and streamlined operations, Smart Curight aims to reduce response times and improve outcomes in critical situations.

Keywords:*Emergency Response, Multispecialty System, Tracking Technology, Real-time Data Analytics, Incident Management, GPS Tracking*

1. INTRODUCTION

In the face of escalating global emergencies and the increasing complexity of modern crises, the demand for advanced, efficient, and reliable emergency response systems has never been greater. Enter Smart Curight, an innovative solution designed to revolutionize emergency management and response through cutting-edge technology and integrated multispecialty capabilities. At its core, Smart Curight is a comprehensive multispecialty emergency response system engineered to address the multifaceted challenges encountered during emergencies, whether they be natural disasters, medical crises, or security threats.

The essence of Smart Curight lies in its ability to seamlessly blend advanced tracking and management solutions with a robust framework of emergency response services. This system is meticulously designed to enhance coordination among various emergency services, streamline communication, and optimize resource allocation in critical situations. By leveraging state-of-the-art technologies such as real-time data analytics, GPS tracking, and artificial intelligence, Smart Curight empowers emergency responders with the tools they need to make informed decisions swiftly and accurately.

One of the standout features of Smart Curight is its multispecialty approach, which integrates diverse emergency response disciplines into a cohesive system. This holistic integration ensures that various aspects of emergency management, including medical, fire, and law enforcement services, can operate in concert, reducing response times and improving overall efficacy. The system's advanced tracking capabilities enable real-time monitoring of emergency situations, allowing for precise location tracking of affected individuals, emergency vehicles, and resources. This real-time data is crucial for dynamic decision-making and efficient management of emergency responses.

In addition to its technical prowess, Smart Curight emphasizes user-friendly interfaces and interoperability. Emergency responders, whether they are paramedics, firefighters, or police officers, can access the system through intuitive platforms that are designed to enhance usability and minimize training time. The system's ability to integrate with existing emergency infrastructure and protocols further amplifies its effectiveness, making it a versatile tool adaptable to various emergency scenarios.

Moreover, Smart Curight is built with a forward-thinking mindset, anticipating future needs and incorporating scalability and flexibility into its design. As emergency situations evolve and new challenges emerge, the system is equipped to adapt and integrate new technologies and methodologies. This adaptability ensures that Smart Curight remains at the forefront of

emergency response innovation, continually enhancing its capabilities to meet the ever-changing demands of the modern world.

Smart Curight represents a significant leap forward in emergency response technology, offering a comprehensive, multispecialty solution that addresses the complexities of contemporary emergencies. Through its advanced tracking, management solutions, and integrated approach, Smart Curight is poised to set a new standard in the field of emergency management, providing a vital tool for responders and ensuring a more effective and coordinated response to emergencies.

1.1 PROJECT DESCRIPTION

Smart Curight is an innovative multispecialty emergency response system designed to revolutionize the way emergency situations are managed and resolved. This advanced system integrates cutting-edge technology to provide real-time tracking, communication, and management solutions tailored for a wide range of emergency scenarios. Its core objective is to enhance the efficiency and effectiveness of emergency response teams, ensuring timely and coordinated actions during critical situations.

Key Features:

Real-Time Tracking and Monitoring: Smart Curight employs sophisticated GPS and IoT technologies to offer real-time tracking of emergency response units, personnel, and equipment. This feature allows for precise location data and movement tracking, ensuring that response teams can be directed to the scene of an emergency with optimal efficiency.

Integrated Communication System: The system includes a robust communication platform that facilitates seamless interaction among emergency responders, control centers, and other stakeholders. It supports voice, video, and data exchanges, ensuring that all parties involved have access to vital information and can collaborate effectively during emergencies.

Advanced Data Management: Smart Curight features a comprehensive data management system that consolidates information from various sources, including incident reports, medical records, and resource inventories. This centralized database enables responders to access critical information quickly and make informed decisions based on real-time data.

Multispecialty Support: The system is designed to cater to a variety of emergency scenarios, including medical, fire, and security incidents. It offers specialized modules and tools for

each type of emergency, ensuring that response teams have the specific resources and protocols they need to handle different situations effectively.

Enhanced Decision-Making Tools: Smart Curight incorporates advanced analytics and decision-support tools that help emergency response teams assess the situation, prioritize tasks, and deploy resources efficiently. These tools provide valuable insights and recommendations based on real-time data, improving the overall response strategy.

User-Friendly Interface: The system features an intuitive and user-friendly interface that simplifies the operation for emergency responders and control center operators. With customizable dashboards and easy-to-navigate controls, users can quickly access essential functions and manage emergency situations with ease.

Benefits:

Improved Response Times: By providing real-time tracking and efficient communication, Smart Curight significantly reduces response times, ensuring that help arrives promptly when it is needed most.

Enhanced Coordination: The integrated communication and data management features facilitate better coordination among various emergency response units, leading to a more organized and effective emergency response.

Better Resource Management: The system's advanced data management and decision-making tools enable optimal allocation and utilization of resources, reducing waste and improving overall efficiency.

Comprehensive Support: With its multispecialty support, Smart Curight ensures that all types of emergencies are handled with the appropriate expertise and resources, enhancing the overall effectiveness of emergency response efforts.

MODLUES

- **Real-Time Tracking Module:** This module utilizes advanced GPS and IoT technologies to provide live tracking of emergency response units, personnel, and critical resources. Key features include:

Location Mapping: Real-time visualization of the location of response teams and resources on a dynamic map.

Movement Tracking: Continuous monitoring of the movement of emergency units to optimize routing and deployment.

Geofencing Alerts: Automated notifications when units enter or exit predefined geographic areas.

Communication and Coordination Module: A robust communication platform that ensures seamless interaction among all stakeholders involved in emergency response. Features include:

Voice and Video Communication: Secure channels for real-time voice and video communication between field units and control centers.

Data Sharing: Ability to exchange incident reports, medical information, and other relevant data instantaneously.

Incident Management Chatrooms: Dedicated chatrooms for different types of emergencies to facilitate focused discussions and decision-making.

Incident Management Module: This module streamlines the management of emergency incidents from start to finish. Key components include:

Incident Reporting: Tools for rapid reporting and documentation of emergency incidents.

Resource Allocation: Automated systems for assigning and tracking resources such as personnel, vehicles, and equipment.

Task Management: Task assignment and tracking features to ensure that all necessary actions are completed efficiently.

Medical Response Module: Specialized tools to support medical emergencies and patient care. Features include:

Medical Records Access: Secure access to patient medical history and emergency contact information.

Emergency Protocols: Predefined protocols for various medical emergencies to guide responders in providing appropriate care.

Telemedicine Support: Integration with telemedicine services for remote consultations and guidance from medical experts.

Fire and Rescue Module: Tailored for fire and rescue operations, this module includes:

Fire Mapping: Real-time heat and fire spread mapping to assist in strategic planning and deployment.

Rescue Operations Support: Tools for coordinating rescue efforts, including building layout information and hazard identification.

Incident Command: Features for establishing and managing an incident command structure to oversee and direct firefighting and rescue operations.

Security Incident Module: Focused on handling security-related emergencies, this module provides:

Threat Detection: Integration with surveillance and threat detection systems to identify and respond to security threats.

Emergency Protocols: Guidelines and checklists for handling various types of security incidents, such as active shooter situations or bomb threats.

Coordination with Law Enforcement: Tools for liaising with law enforcement agencies and coordinating joint response efforts.

1.2 COMPANY PROFILE:

Colan Infotech is a leading technology solutions provider specializing in IT consulting, software development, and digital transformation services. With a strong focus on leveraging cutting-edge technologies and innovative strategies, Colan Infotech delivers tailored solutions to meet the unique needs of businesses across various industries. Their expertise spans across custom software development, enterprise application integration, cloud services, and data analytics, enabling clients to optimize operations, enhance efficiency, and drive growth. Committed to excellence and client satisfaction, Colan Infotech combines technical prowess with a deep understanding of industry trends to deliver impactful solutions that empower organizations to achieve their digital goals.

2. LITERATURE SURVEY

2.1 EXISTING AND PROPOSED SYSTEM

Existing System

Current Emergency Response Systems

In today's landscape, emergency response systems are typically segmented into various specialized units, each addressing different aspects of emergency management such as medical emergencies, fire, police, and disaster response. These systems often function in isolation, leading to challenges in coordination and information sharing among different response teams. Common features of existing systems include:

Dispatch and Communication: Traditional emergency response systems rely on centralized dispatch centers to manage calls and coordinate resources. Communication channels are usually radio-based or telephone-based, with limited integration between different agencies.

Manual Tracking and Coordination: Emergency units often use manual logs or outdated tracking systems to monitor the status of resources and response activities. This can lead to delays in response times and difficulties in managing resources effectively.

Limited Data Integration: Existing systems may have limited capability to integrate data from various sources, such as incident reports, GPS tracking, and real-time updates from field units. This lack of integration can impede the overall efficiency and effectiveness of emergency response operations.

Standardized Protocols: Many systems operate based on standardized protocols and procedures that may not always be adaptable to specific or evolving emergency scenarios. This can limit the flexibility and responsiveness of the system in dynamic situations.

Proposed System

Smart Curight: A Comprehensive Multispecialty Emergency Response System

The proposed Smart Curight system aims to revolutionize emergency response through advanced tracking and management solutions designed for enhanced efficiency, coordination, and adaptability. Key features of the Smart Curight system include:

Integrated Multispecialty Dashboard: Smart Curight offers a unified dashboard that integrates data from various emergency response units including medical, fire, police, and disaster management. This centralizes information and streamlines coordination among different agencies, facilitating a more cohesive response strategy.

Real-Time Tracking and Monitoring: Utilizing cutting-edge GPS and IoT technologies, Smart Curight provides real-time tracking of emergency units and resources. This enables dynamic

resource allocation based on the current status of each unit and the evolving nature of the emergency situation.

Advanced Communication Network: The system incorporates an advanced communication network that supports multiple channels including voice, video, and data transfer. This ensures seamless communication between field units and command centers, reducing the likelihood of miscommunication and delays.

Data Analytics and Predictive Modeling: Smart Curight employs data analytics and predictive modeling to anticipate potential emergencies and optimize response strategies. By analyzing historical data and real-time inputs, the system can forecast trends and provide actionable insights for proactive emergency management.

Adaptive Protocols and Decision Support: Unlike traditional systems, Smart Curight features adaptive protocols that can be customized based on specific emergency scenarios. The decision support system offers recommendations and best practices based on real-time data and predefined criteria, enhancing the flexibility and effectiveness of response efforts.

User-Friendly Interface and Mobile Access: Designed with usability in mind, Smart Curight includes a user-friendly interface accessible via mobile devices. This allows field personnel to access critical information, update status, and communicate effectively from any location, further improving response times and coordination.

Comprehensive Reporting and Analysis: The system generates detailed reports and analyses of emergency responses, providing valuable insights for post-incident reviews and continuous improvement. This data-driven approach helps in identifying areas for enhancement and ensures that lessons learned are integrated into future response strategies.

2.2 FEASIBILITY STUDY



The "Smart Curight" system proposes a cutting-edge approach to emergency response, focusing on integrating advanced tracking and management solutions to enhance efficiency and effectiveness in emergency situations. The core feasibility of this project hinges on several key factors including technological viability, market demand, and operational practicality.

Technological Viability

"Smart Curight" leverages state-of-the-art technologies such as real-time tracking, advanced communication systems, and data analytics to streamline emergency response operations. The feasibility of this system is supported by the availability of robust technologies that can be integrated into a single, cohesive platform. Technologies like GPS tracking, AI-driven decision-making algorithms, and IoT sensors are already established and have proven reliability in various applications, which underpins the technical feasibility of implementing such a system for emergency management.

Market Demand

There is a significant and growing demand for more efficient and responsive emergency management solutions due to increasing urbanization and the complexity of modern emergencies. The need for multispecialty systems that can address a wide range of emergency scenarios—from medical crises to natural disasters—drives the market potential for "Smart Curight." Research indicates a rising preference among municipalities, healthcare providers, and emergency services for integrated solutions that can improve response times and coordination.

Operational Practicality

The practical implementation of "Smart Curight" involves evaluating the readiness of existing infrastructure, the integration of new technologies with current systems, and the training requirements for personnel. Operational feasibility will be determined by assessing the compatibility of new technologies with existing emergency response frameworks and ensuring that the system can be effectively deployed and maintained. Pilot programs and phased rollouts can help mitigate risks and address any operational challenges before full-scale implementation.

2.3 TOOLS AND TECHNOLOGIES

JAVA

The development of Android applications requires a specific set of software tools and environments to ensure compatibility and efficiency. The primary programming languages used are Java and XML. Java is the core language for writing the application logic, thanks to its robustness and versatility. It allows for the creation of complex, reliable, and maintainable code. XML, on the other hand, is used for designing and configuring the user interface and layout of the application. This separation of logic and presentation helps maintain a clear structure in the application development process.

For development, the operating system should be Windows 8 or above. This version of Windows provides the necessary support and compatibility for running development tools and IDEs effectively. Additionally, the Android operating system version 3.0 or higher is required for deploying the applications. This ensures that the applications are compatible with the modern features and functionalities of the Android platform.

The development environment is set up using Android Studio, which is the official Integrated Development Environment (IDE) for Android development. Android Studio offers a comprehensive suite of tools and features, including a code editor, a robust emulator, and a variety of debugging and performance analysis tools. It is designed to streamline the development process, making it easier to build, test, and deploy Android applications. The combination of Java for programming, XML for user interface design, and Android Studio for development provides a powerful and efficient framework for creating high-quality Android applications.

2.4 Hardware and Software Requirements

2.4.1 Hardware Requirements

Processor: Dual Core 3.0 GHz

Hard Disk: 160 GB

RAM: 3 GB

Android Device: Smartphone or Tablet

2.4.2 Software Requirements

Programming Languages: Java, XML

Operating System: Windows 8 or above for development, Android 3.0 or above for deployment

Development IDE: Android Studio

3. SOFTWARE REQUIREMENT SPECIFICATION

3.1 USERS

3.1.1 Doctor Search : This feature allows users to view detailed profiles of doctors with a single request. It simplifies the process of finding the right healthcare professional by providing comprehensive information.

3.1.2 Login/Register: Users must register with valid information. Upon successful registration, they can log in using their mobile number and a One-Time Password (OTP) for authentication.

3.1.3 Find and Book : Users can search for doctors' advice or book appointments for check-ups and treatments. Additionally, the application enables patients to search for relevant tests and schedule appointments for those tests.

3.1.4 Reports : Patients can upload their prescriptions, and doctors will provide treatment recommendations based on the medical conditions indicated in the reports. The application maintains a record of uploaded reports and allows patients to select images for prescription uploads.

3.1.5 Billing: This module handles financial management by ensuring accurate billing. It includes processes for handling claims efficiently and maintaining the application's services.

3.1.6 Patient Communication : The application facilitates communication by generating and distributing balance statements to patients. It improves the patient experience and supports effective interactions between doctors and patients.

3.1.7 Treatment : The application provides a facility for patients to receive appropriate treatment based on their conditions and schedule laboratory tests.

3.1.8 Prescription : The application supports both image and digital prescriptions, making it easier for patients to maintain their medical records.

3.2 FUNCTIONAL REQUIREMENTS

Overall Description

3.2.1 Product Perspective

Smart Curight will be a web-based and mobile application system designed to interface with existing emergency response infrastructure. It will integrate with various data sources, including GPS, communication systems, and health records.

3.2.2 Product Functions

Real-time Tracking: Track emergency vehicles and personnel in real-time.

Automated Dispatching: Automatically dispatch emergency units based on location and type of emergency.

Incident Management: Record and manage incidents, including detailed logs and status updates.

Multispecialty Coordination: Facilitate coordination among medical, fire, police, and disaster management teams.

Data Integration: Integrate with GIS for mapping and routing, and with other databases for relevant information.

3.2.3 User Classes and Characteristics

Emergency Dispatchers: Use the system to dispatch units, track responses, and manage incidents.

Emergency Responders: Access real-time information and updates about incidents and assignments.

Administrators: Manage user accounts, system settings, and overall system configuration.

End Users (Citizens): Report emergencies and track response progress via a public-facing interface.

3.2.4 Operating Environment

Web Application: Compatible with modern web browsers (Chrome, Firefox, Edge).

Mobile Application: Available for iOS and Android platforms.

Server Environment: Hosted on secure cloud infrastructure with redundancy and failover capabilities.

3.2.5 Design and Implementation Constraints

Compliance: Must comply with local regulations and standards for emergency response.

Security: Ensure data protection and privacy according to relevant data protection laws.

Scalability: The system must handle varying loads, from small-scale incidents to large-scale emergencies.

3.2.6 User Documentation

User Manuals: Detailed guides for each user role.

Training Materials: Tutorials and training sessions for new users.

Support: Helpdesk and technical support resources.

3.3 Non-Functional Requirements

Performance Requirements (3.1.1): The application should offer high performance with fast speed and responsiveness. Technologies such as Android, XML, and Retrofit should be optimized to ensure quick response times to user interactions.

Portability (3.1.2): The application should be compatible with mobile systems and portable across various devices. The development team must use technologies that enhance portability and ensure broad user acceptance.

Security (3.1.3): Security is a priority for the application. It must feature a secure login system, maintain report confidentiality, and use OTP for user authentication. Each module should include registration protocols to ensure comprehensive security.

Usability (3.1.4): The application should be user-friendly and accessible. It should allow patients to easily book appointments, track health parameters (such as blood pressure, BMI, and sugar levels), and receive medications online. Users should also have access to articles or tips from doctors.

Reliability (3.1.5): The application must be reliable and function consistently. It should be built using technologies that are regularly updated and improved, ensuring trustworthy performance. Users should find the application dependable and easy to use.

Maintainability (3.1.6): The application should be easy to maintain and well-organized. Patient, doctor, and payment details should be securely stored in a database like Firebase. Any modifications to the application should not disrupt its functionality, and administrators should be able to manage all relevant details efficiently.

4.SYSTEM DESIGN

4.1 System Perspective

The system perspective focuses on the overall design and integration of the application. The design approach employed involves a circular structure, which facilitates the seamless integration of all application features. This design method ensures that different trade streams are interconnected, thereby enhancing the system's operational efficiency. By structuring the system in this manner, we are able to validate the practical application of the clinical meetings more effectively. The circular design allows for the creation of a unified system, where each section contributes to the overall functionality and integration of the complete system.

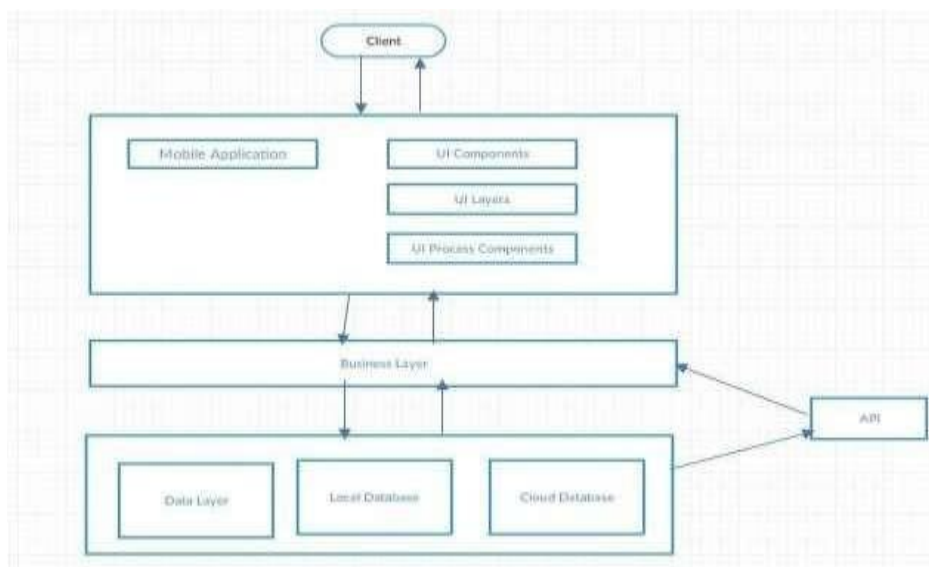


Fig:4.1.1

4.2 Context Diagram

A Data Flow Diagram (DFD) begins with a Context Diagram, which provides an overview of the entire system as a single process and illustrates its interactions with external entities such as users, other systems, or databases. The Context Diagram offers a high-level view of the system and its external connections, presenting a broad perspective on how the system integrates with the external environment.

As the project progresses, the DFD is expanded into more detailed levels. Each subsequent level delves deeper into the system's processes and data flows. Lower-level diagrams offer a more granular view, detailing individual processes and illustrating how data moves between them. This hierarchical approach allows for a comprehensive understanding of the system's internal workings and its interactions with external elements.

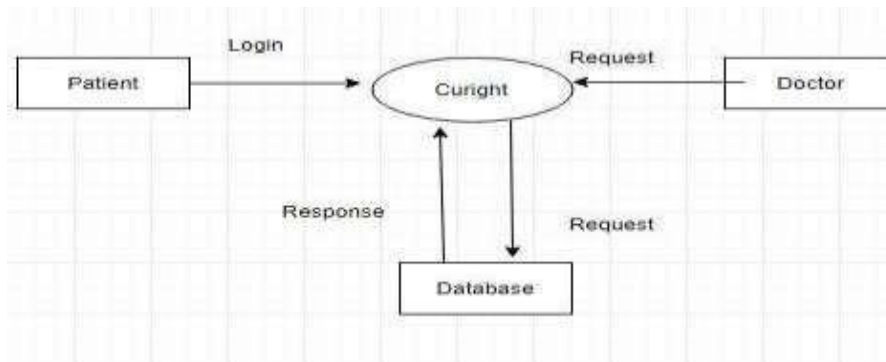


Fig4.2.1 DFD Level 1

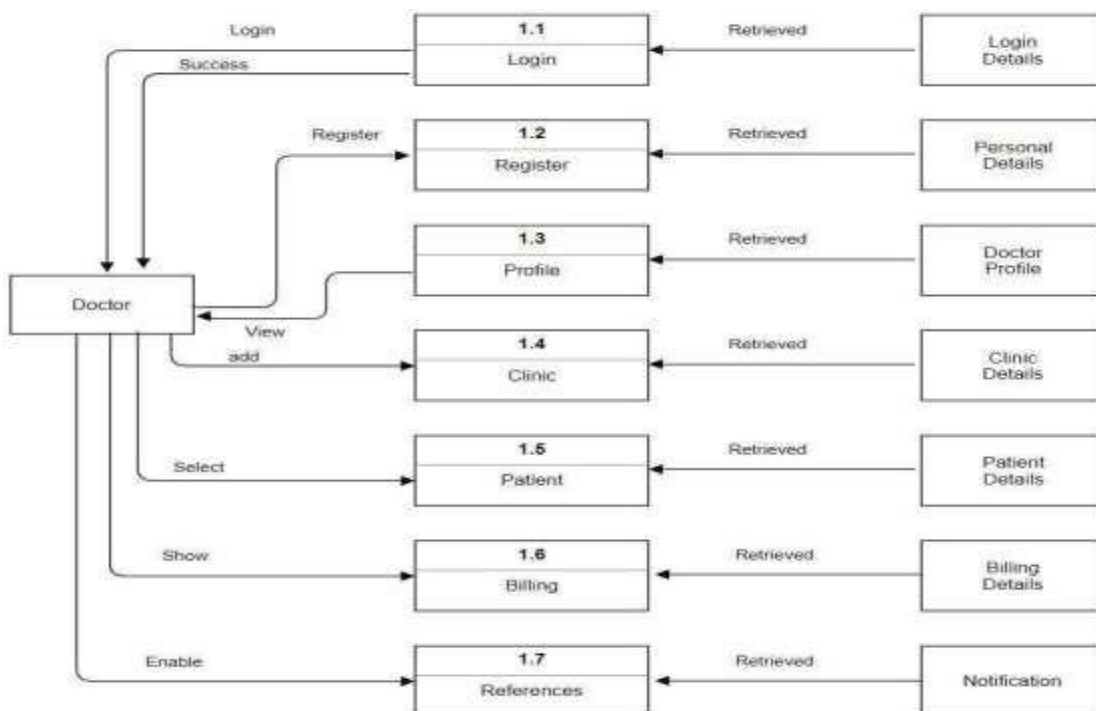


Fig:4.2.2 DFD2

5.Detailed Design

Detailed design is a pivotal phase in the software development lifecycle, where the high-level requirements established in earlier stages are transformed into a comprehensive and structured plan for implementation. This phase is crucial as it lays out a precise blueprint for building the final product.

Key Aspects of Detailed Design:

Initial Requirements: The process begins with a thorough understanding of the system's requirements, which were gathered during the analysis and design phases. These requirements form the foundation of the detailed design, ensuring that the system will fulfill the intended functionalities and meet user needs.

The detailed design phase focuses on translating these requirements into specific, actionable components, defining how each part of the system will function and interact. This meticulous planning is essential for guiding the development process and ensuring that the final product aligns with the initial vision and objectives.

5.1 USE CASE DIAGRAM

Doctor:

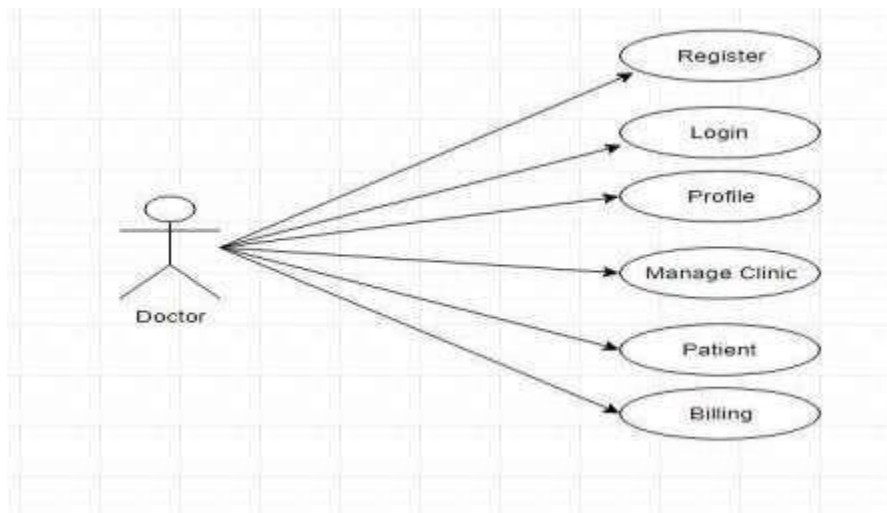


Fig:5.1.1 Doctor Use case

The "Doctor" use case outlines the primary roles and responsibilities of doctors within the system, serving as the central hub that connects various features and functionalities. This use case includes essential interactions such as login, registration, and profile management, as well as specific tasks related to clinic management, billing, and patient interactions. It provides a comprehensive view of how doctors engage with the system and the various functionalities they utilize.

When a doctor logs in or registers, they access a personalized profile where they can update and manage their information, including contact details, specialization, and qualifications. The profile management feature allows doctors to keep their information accurate and current within the system.

Beyond individual profile management, doctors can efficiently manage their clinics through the system. This includes scheduling and organizing appointments, tracking patient visits, and overseeing clinic availability. By integrating these clinic management functionalities, the system helps streamline the workflow for doctors, enhancing their productivity and overall efficiency.

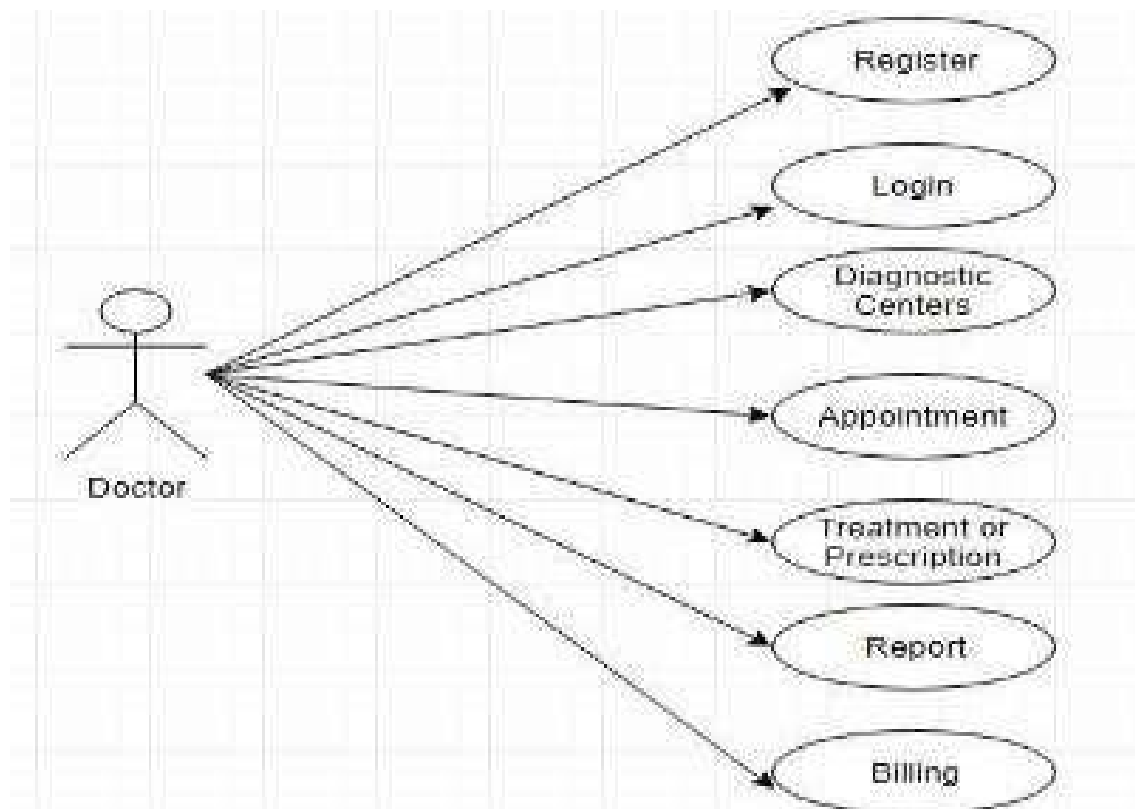


Fig:5.1.2 Patient

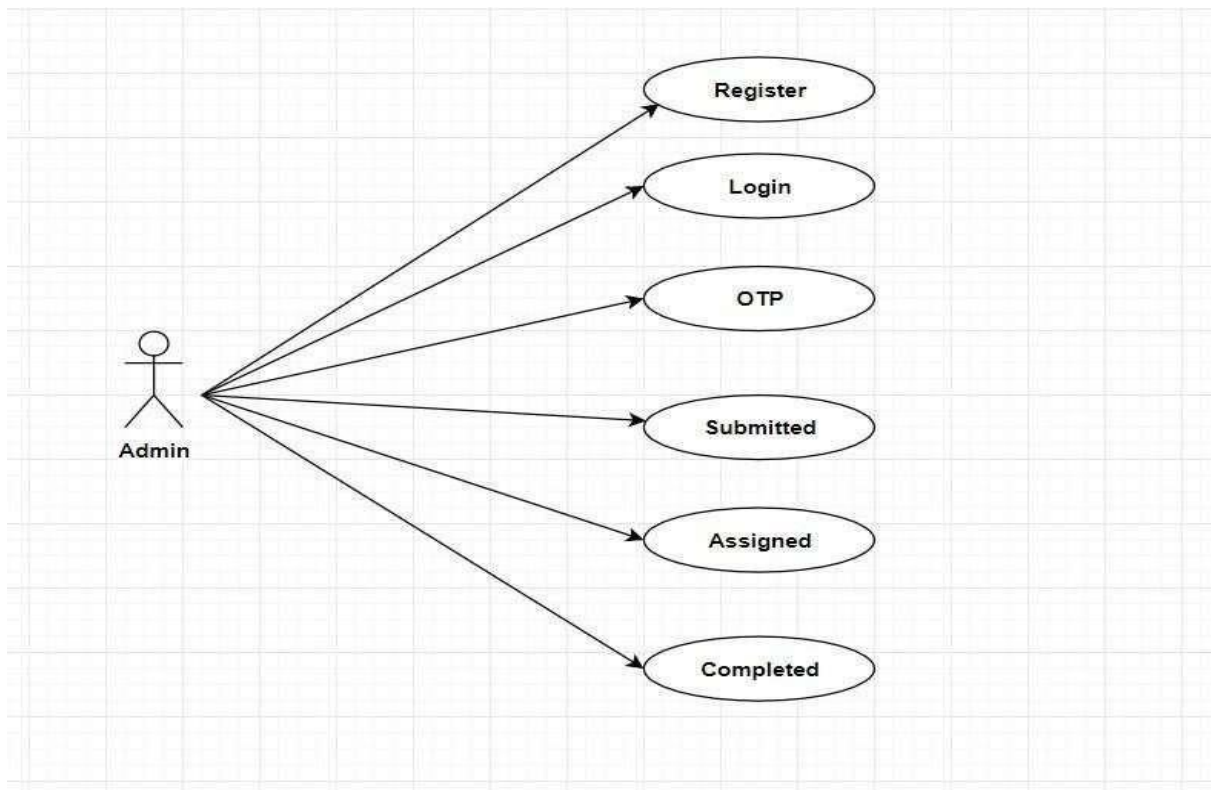


Fig:5.1.3 Admin

5.2 Sequence Diagrams

Doctors:

Login and Select Appointment Date:

The doctor starts by sending a login request to the system.

The system validates the doctor's credentials by checking them against the database.

Upon successful validation, the system presents an interface for the doctor to select an appointment date.

The doctor selects the desired appointment date through the system.

Upload Documents and Select Appointment Slot:

After choosing the appointment date, the doctor requests to upload any necessary documents related to the appointment.

The system receives the document and stores it in the database.

The doctor then selects an available appointment slot from the system.

The system updates the appointment slot availability in the database.

Manage Another Clinic:

The doctor may request to manage another clinic.

The system processes this request and updates the relevant information accordingly.

Update Patient Status:

During an appointment, the doctor examines the patient and may need to update the patient's status in the system.

The doctor sends a request to update the patient's status.

The system validates the request and updates the patient status in the database.

Upload Prescription:

If needed, the doctor can upload a prescription for the patient.

The prescription is sent to the system, which then stores it in the database.

The system also records the fee details associated with the appointment and updates the prescription record.

Validation and Display of Doctor's Dashboard:

To access the doctor's dashboard, the system performs an additional validation of the doctor's identity.

Once verified, the system retrieves pertinent information from the database.

The doctor's dashboard is then displayed with updated information, including fee details, appointment information, and patient status.

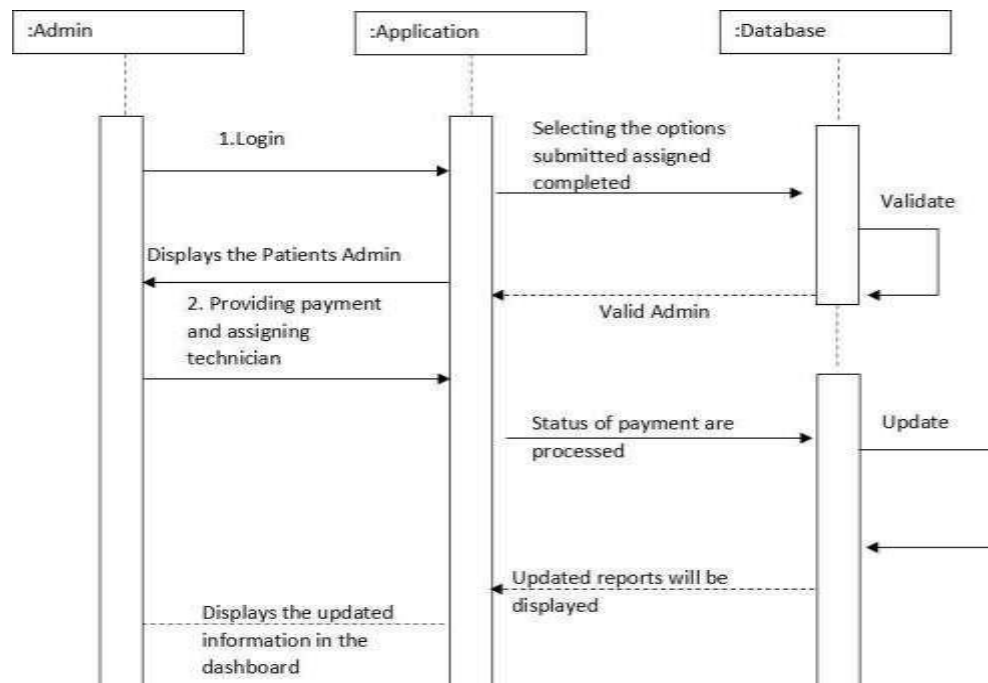


Fig:5.2.1 Sequence Diagram

5.3 ACTIVITY DIAGRAMS

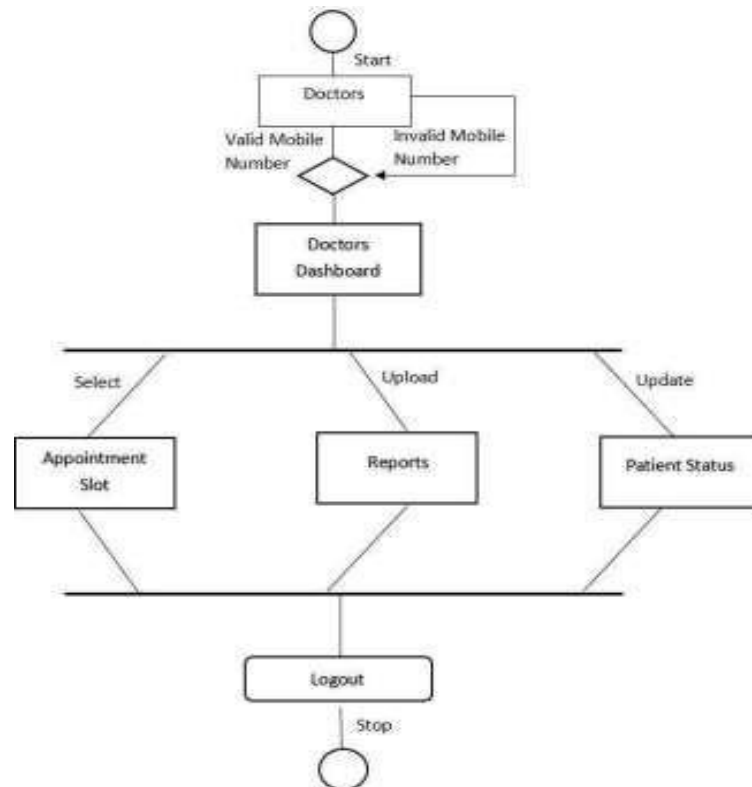


Fig 5.3.1 Doctor

Patient:

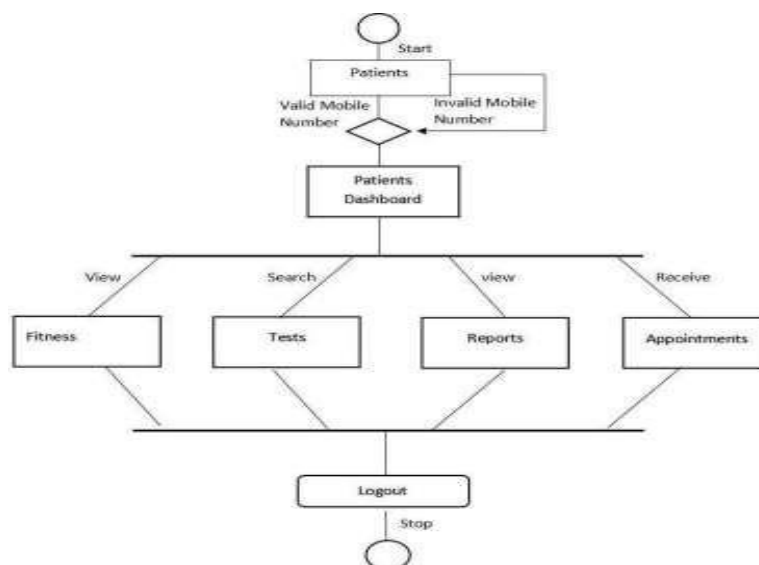
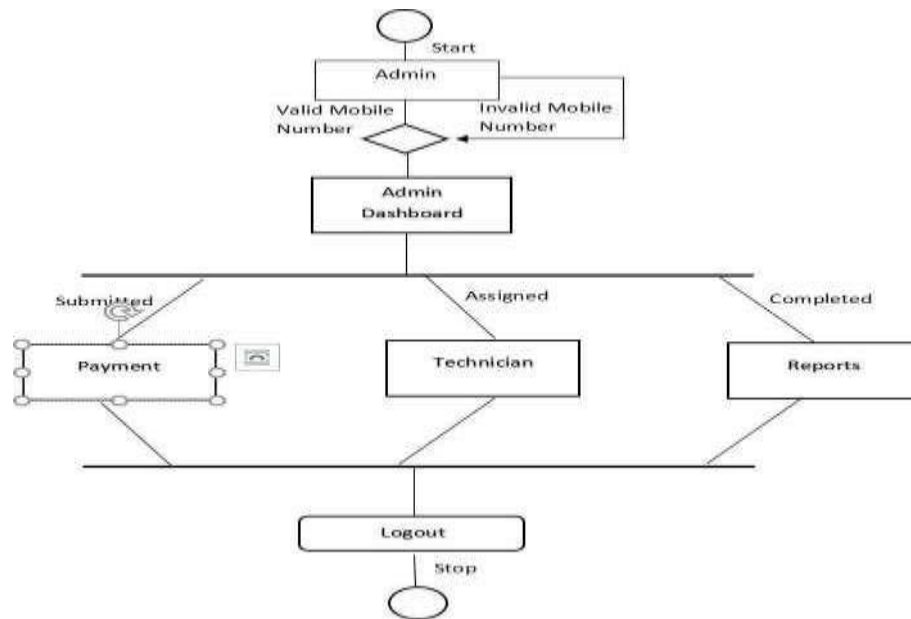


Fig 5.3.2 Admin



5.4 CLASS DIAGRAM

This class represents an arrangement, which may involve various details not mentioned in the description. In the context of the expert class, this represents the arrangements that an expert will deliver. However, additional details regarding the arrangement would need to be specified.

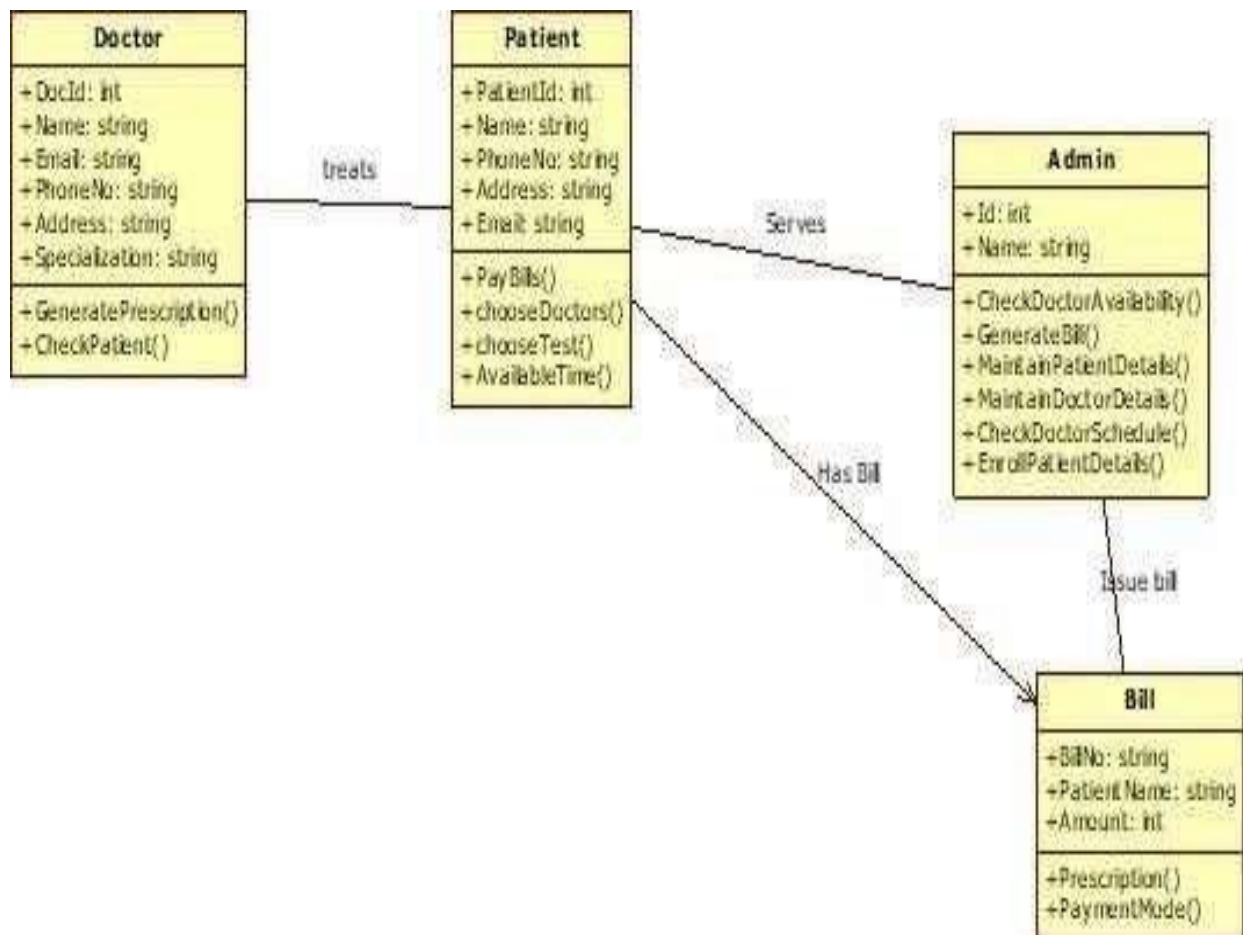


Fig:5.4.1 Class diagram

5.5 ER DIAGRAM

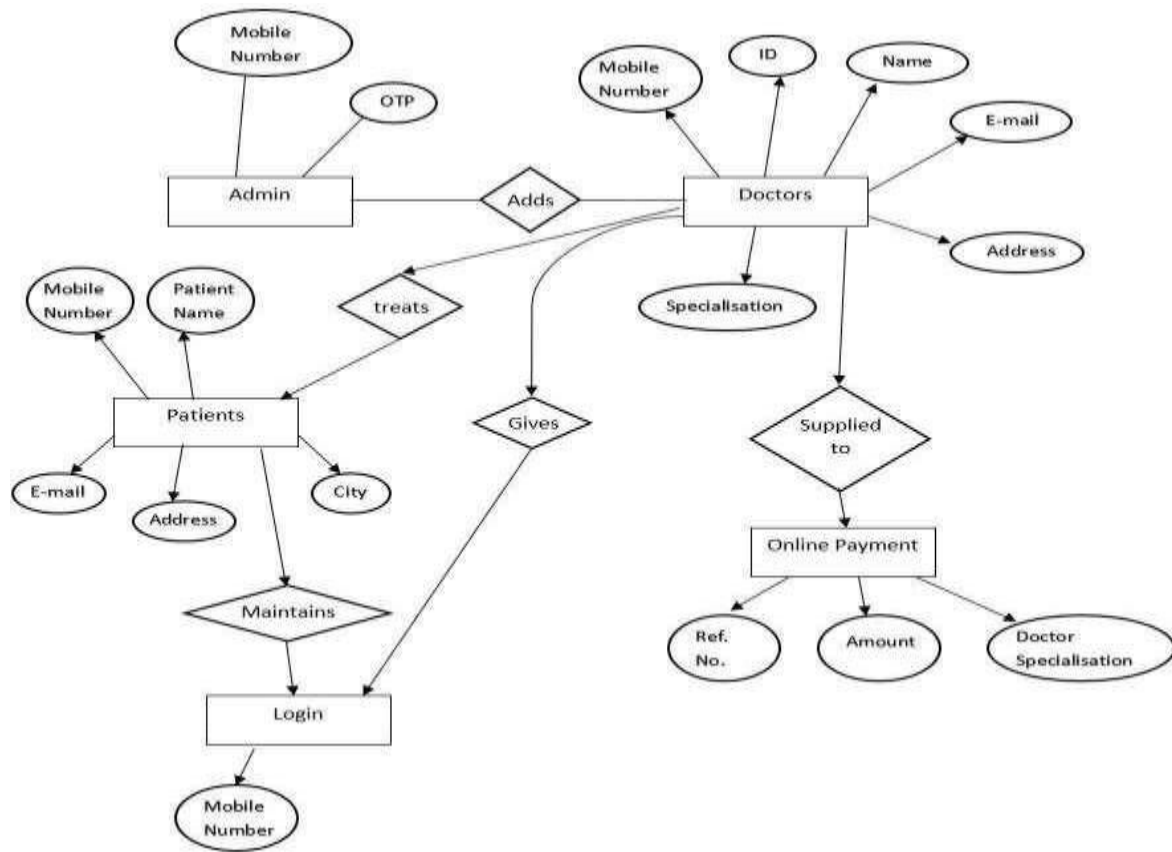


Fig:5.5.1 ER diagram

6.IMPLEMENTATION

6.1 SCREENSHOTS

6.1.1 Registration

The screenshot shows a registration form for 'curright'. At the top is the 'curright' logo with a heart rate line above it. Below the logo is the word 'Profile' and a blue circular icon with a white pencil. The form contains five input fields: 'Name' with the text 'mtee', 'Email' with 'meeqese@gmail.com', 'Mobile Number' with '9707897870', and 'Specialization' with 'General Physician'. A large blue 'SUBMIT' button is at the bottom.

curright

Profile



Name
mtee

Email
meeqese@gmail.com

Mobile Number
9707897870

Specialization
General Physician

SUBMIT

Fig 6.1.1 Register

Once the registration is complete, the user is redirected to the login screen.

The login screen usually requires the user to enter their mobile number (or possibly email) and a password to access their account

6.1.2 Patient Dashboard

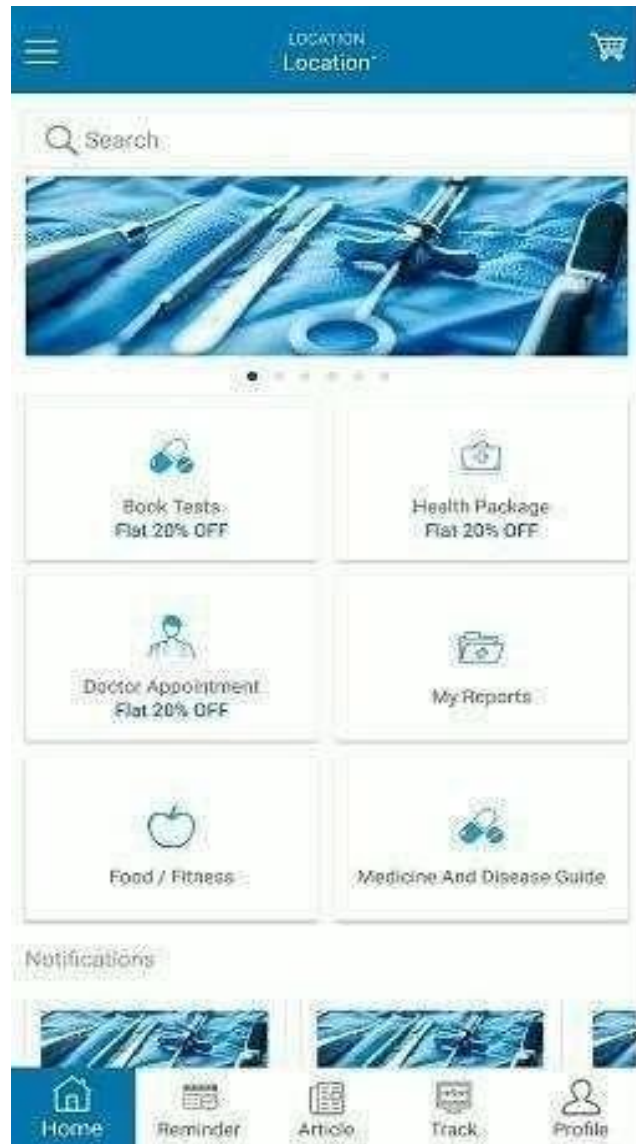


Fig 6.1.2 Patient Dashboard

actual implementation and design of these features will depend on the specific requirements and goals of the application. Additionally, ensuring patient data privacy and security should be a top priority throughout the application development process

6.1.3 Diagnostic centres



Fig 6.1.3 Diagnostic centres

The specific design and layout of the "Book Tests" page will depend on the application's user interface and user experience considerations. The goal is to make the booking process easy to understand, visually appealing, and user-friendly while providing all the necessary information for clients to make informed decisions about their medical tests..

6.1.4 Food/Fitness

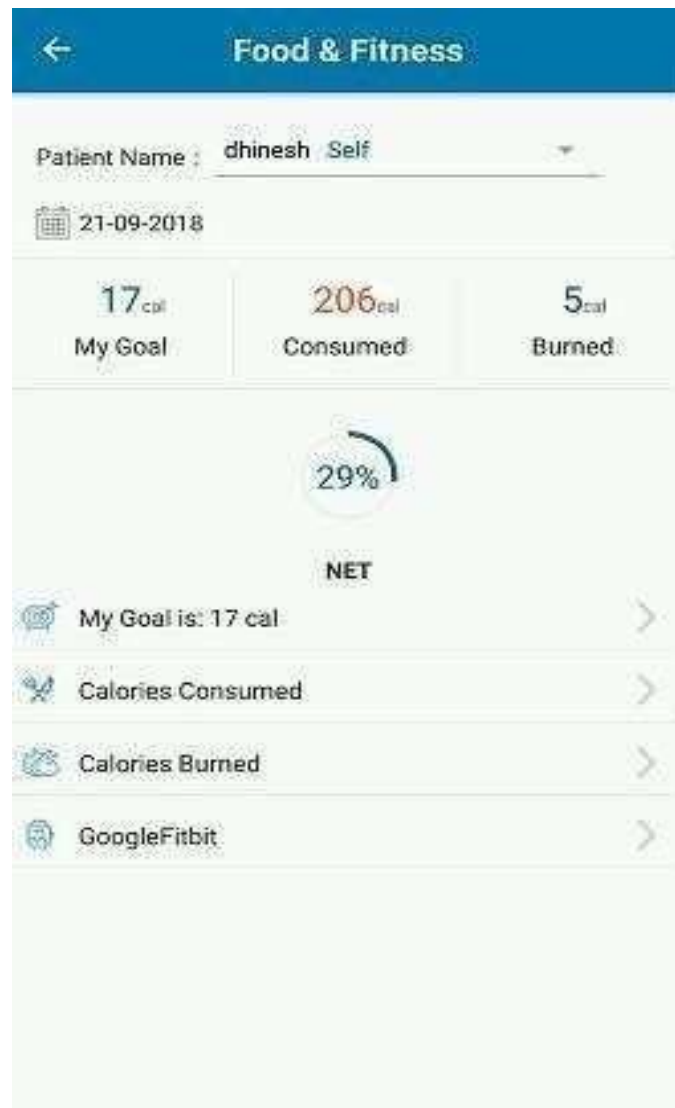


Fig 6.1.4 Food

The overall goal is to provide clients with a comprehensive and user-friendly health and fitness tracking experience, combining food intake data with fitness details from Google Fitbit to offer valuable insights and support their health objectives

6.1.5 Article

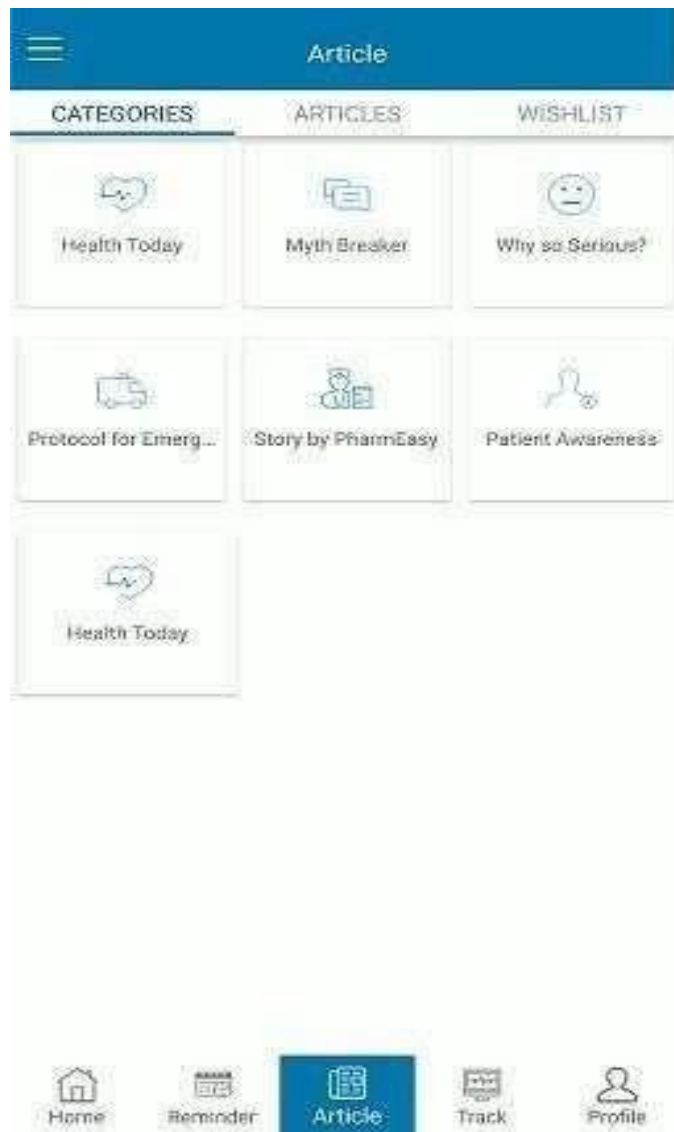


Fig 6.1.5 Article

feature is to provide clients with valuable knowledge and resources, foster a supportive health community, and help them make informed decisions to improve their well-being. By combining expert insights with user-generated content and sharing options, the application encourages collaboration and engagement among its users

6.1.6 BMI track

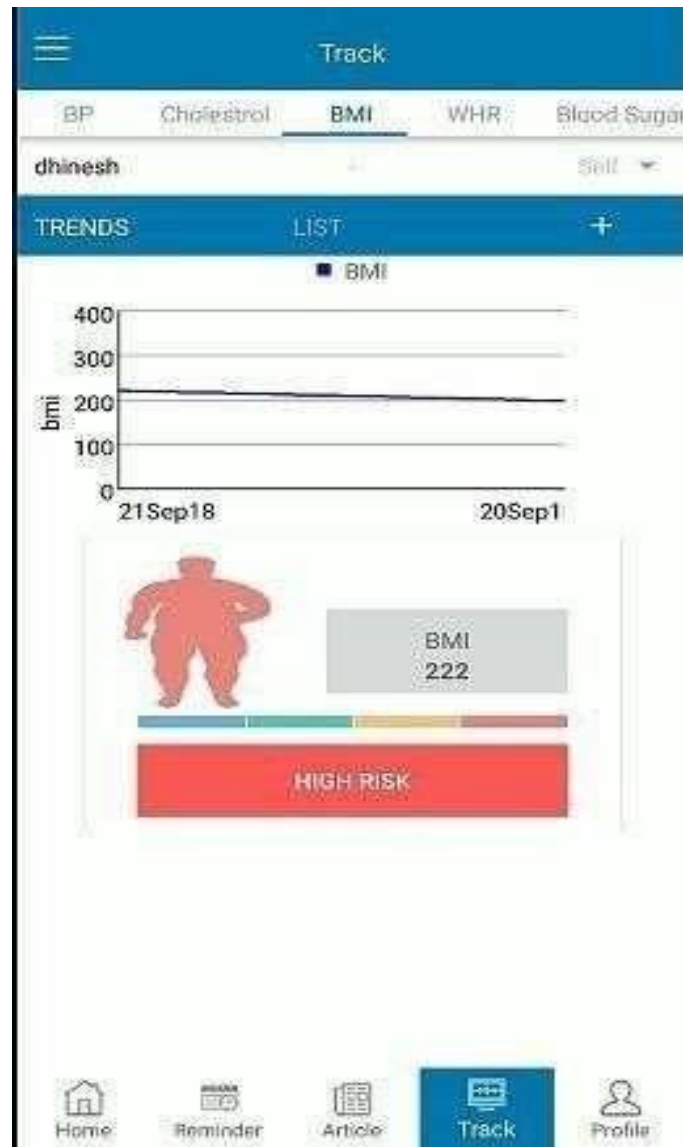


Fig 6.1.6 BMI

The BMI tracking module is a valuable tool to help patients understand their body composition and make informed decisions regarding their health and lifestyle. By offering a quantifiable chart and considering waist and hip measurements, the application provides a more comprehensive assessment of body fat distribution and potential health risk.

6.1.7 Profile

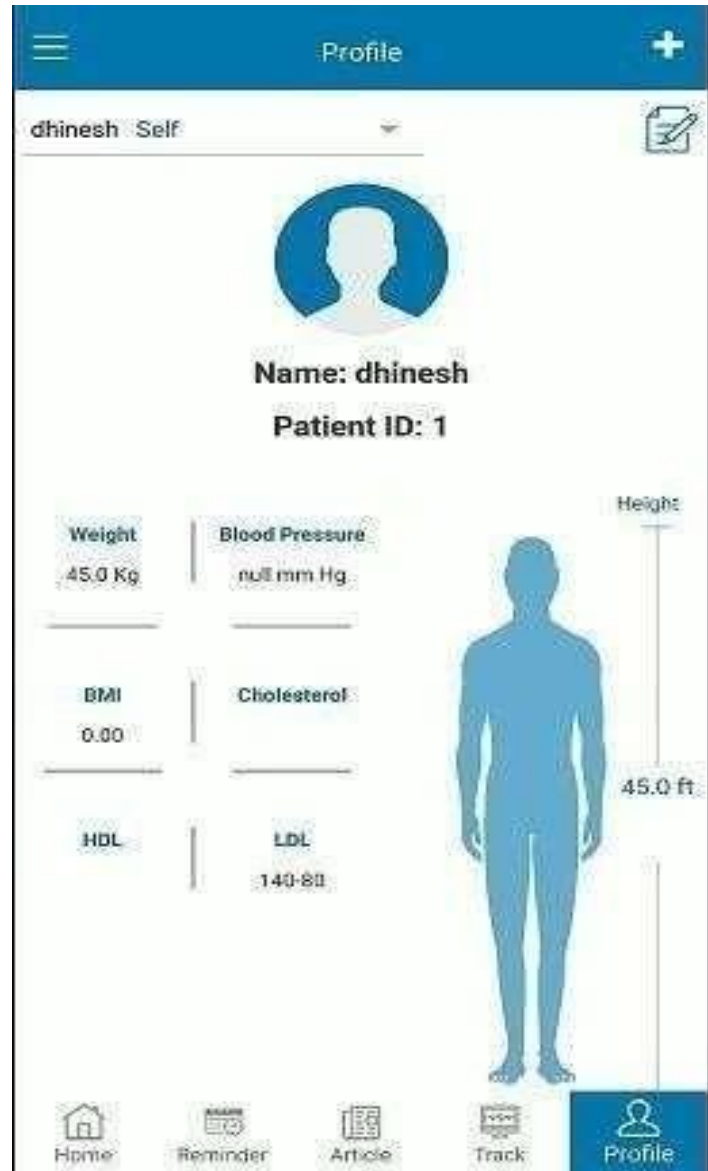


Fig 6.1.7 Profile

The patient's profile is designed to provide a comprehensive overview of their health status and progress towards their health goals. It serves as a valuable tool for patients and healthcare professionals to track and manage health-related data efficiently. Privacy and security measures should be implemented to safeguard sensitive health information within the patient's profile.

6.1.8 Doctor Dashboard



Fig 6.1.8 Doctor Dashboard

The expert's dashboard is designed to provide a central hub for managing their interactions with patients, offering educational content, and fostering professional growth within the platform. It streamlines their workflow, allowing them to focus on providing high-quality care and valuable educational resources

6.1.9 Manage clinic

Manage Clinic

Clinic name
test clinic

Clinic address
test clinic address

Phone number
9999999999

Enter Slot Timing in Minutes
30

Available Days in Week

☒	Monday	From - 21:47	To - 14:47
☐	Tuesday	From	To
☐	Wednesday	From	To
☐	Thursday	From	To
☐	Friday	From	To
☐	Saturday	From	To
☐	Sunday	From	To

SAVE

Home Calendar Patient Preference

Fig 6.1.9 Manage Clinic

By providing the option for experts to manage multiple office locations and specifying their availability, the application enables clients to schedule appointments at their preferred office and helps experts efficiently manage their consultations across various centers. This feature enhances flexibility and accessibility for both experts and clients.

6.1.10 Basic Information

The screenshot displays a mobile application interface for a patient's profile. At the top, a blue header bar contains a menu icon, the word "Patient", and a notification bell icon. Below the header, the patient's name "Tabbasum Nargis" is shown next to a profile picture icon, with the text "Last visit : 08-04-2018" underneath. A horizontal tab bar below the name includes "PROFILE", "NOTES", "PRESCRIPTION", and "REPORTS". The "PROFILE" tab is selected, leading to a section titled "Basic Info". This section contains a table with the following data:

Age	2.6 Years
Gender	Male
Pressure	80/120
Smoking	25
Diabetes	20
BMI	30
Weight	82 Kg

Below the table is a blue circular button with a white plus sign. At the bottom of the screen is a navigation bar with four icons and labels: "Home", "Calendar", "Patient", and "Preference".

Fig:6.1.10

Important to prioritize data privacy and ensure that patients have control over their personal health information. Patient profiles are valuable tools for healthcare providers, helping them provide individualized care and tailored treatment plans based on the patient's health history and current health status

6.1.11 Notes

The screenshot displays the 'Patient' app interface. At the top, there is a blue header with a menu icon on the left, the word 'Patient' in the center, and a bell icon on the right. Below the header, a patient profile section shows a circular profile picture, the name 'Tabbasum Nargis', and the text 'Last visit : 08-04-2018'. A horizontal tab bar below the profile section has four tabs: 'PROFILE', 'NOTES' (which is selected and underlined), 'PRESCRIPTION', and 'REPORTS'. The 'NOTES' section contains a list of notes. Each note entry consists of a grey header with a label (e.g., 'test for note', 'gffc', 'dbbdb', 'yby', 'yby', 'ubub') and a numerical value '0' on the right. Below each header is a white input field containing the same text as the header. A blue circular button with a white plus sign is positioned to the right of the 'ubub' note. At the bottom of the screen is a navigation bar with four icons and labels: 'Home' (house icon), 'Calendar' (calendar icon), 'Patient' (person icon), and 'Preference' (gear icon).

Fig 6.1.11 Notes

By providing a platform for clients to add notes about their medical tests and offering a comprehensive test overview with relevant disease information and treatment recommendations, the application supports clients in managing their health and facilitates effective communication with healthcare providers. The application can become a valuable tool in the client's healthcare journey, fostering better understanding and decision-making regarding their health conditions and treatment options..

6.1.12 Prescription

The screenshot displays a mobile application interface for managing patient prescriptions. The top section, titled 'Patient', contains a 'Prescription 1' entry. This entry includes a frequency field set to '1 - 0 - 1', a duration field set to '10 Weeks', and a notes field with the text 'Before Meals - test instruction'. Below the notes field is a button labeled '+ Add another'. At the bottom of the form is a prominent blue 'Save Prescription' button. The application's navigation bar at the very bottom features four icons: a house for 'Home', a calendar for 'Calendar', a person for 'Patient', and a gear for 'Preference'.

Fig 6.1.12 Prescription

By offering a user-friendly interface that presents specialist solutions, treatment plan details with remedy frequency and duration, and predefined notes, the application helps clients better understand their treatment options and follow the prescribed regimen effectively. This feature promotes collaborative care between specialists and clients, ensuring a more comprehensive and personalized healthcare experience..

6.1.13 Payment Process

←

Payment Details

dhinesh

Registration No : 123

Phone No : 9940821681

Email ID : dhinesh@innovellent.com

Referral Doctor :

Doctor Contact No :

Address : 11-B Ganga Ram Hospital Marg

Test Name	Amount
Thyroid	1000

Total Payment Required

1000

Select Discount Code

Enter Code

Total Payable Amount

1000

Mode

Cash

Online

PAY NOW

Fig 6.1.13 Payment

6.1.14 Reports



Fig 6.1.14 Report

Clients can add the reports with the goal that the reports will be seen in my reports.

7.SOFTWARE TESTING

Programming testing is to be sure a system for recognizing and revising bugs and blunders in a web application. It guarantees that the application works appropriately and meets the predetermined necessities prior to being conveyed to clients.

There are two significant strategies for programming testing: discovery testing and white box testing. The two strategies assume a critical part in the product improvement process:

Discovery Testing: This testing strategy includes testing the application disregarding its inside code structure. Analyzers center around the application's bits of feedbacks and results to confirm the event that it acts true to form. Black box testing is valuable for assessing the application's usefulness according to the client's point of view.

White Box Testing: Otherwise called underlying testing or code-based testing, this technique includes testing the inside code of the application. Analyzers look at the application's code to guarantee that all branches and ways are appropriately executed, distinguishing legitimate mistakes, and surveying code inclusion. White box testing guarantees the rightness of the application's inside rationale.

Testing assumes a basic part in the product improvement process. A clear cut testing plan frames the objectives and techniques to accomplish those objectives. Each period of the arrangement includes execution and checking to guarantee progress is lined up with the arrangement and wanted results are accomplished.

7.2.1 Registration Screen

Scenario 1:

Action: Entering a valid name (Doctor)

Result: Name is known, and the sign-in screen is displayed.

Scenario 2:
Action: Entering an invalid name (Doctor - Unknown Name)

Result: The sign-in screen will be terminated, and the sign-in screen is not displayed.

Scenario 3:

Action: Entering a valid email (Doctor)

Result: Email is known, and the sign-in screen is displayed.

Scenario 4:
Action: Entering an invalid email (Doctor - Unknown Email)

Result: The sign-in screen will be terminated, and the sign-in screen is not displayed.

Scenario 5:

Action: Entering a valid specialization (Doctor)

Result: Specialization is known, and the sign-in screen is displayed.

Scenario 6:

Action: Entering an invalid specialization (Doctor - Unknown specialization)

Result: The sign-in screen will be terminated, and the sign-in screen is not displayed.

7.2.2 Sign in Screen

Scenario 1:

Action: Entering a valid mobile number (Doctor)

Result: Mobile number is known, and the Doctor's Dashboard is displayed.

Scenario 2:

Action: Entering an invalid mobile number (Doctor - Unknown Mobile Number)

Result: The Doctor's Dashboard will be terminated, and the Doctor's Dashboard is not displayed.

7.2.1 Payment

Screen

Action: Entering a valid Registration number (Patient)

Result: Registration number is known, and the user's screen is displayed.

Scenario 2:

Action: Entering an invalid Registration number (Patient - Unknown Registration number)

Result: The user's screen will be terminated, and the user's screen is not displayed.

Scenario 3:

Action: Entering a valid test name & amount (Patient)

Result: Test is known, and the amount is known, and the user's screen is displayed.

Scenario 4:

Action: Entering an invalid test name & amount (Patient - Unknown Test & Unknown Amount)

Result: The user's screen will be terminated, and the user's screen is not displayed.

Scenario 5:

Action: Enter a valid payable amount (Patient)

Result: Payable amount is known, and the user's screen is displayed.

Scenario 6:

Action: Entering an invalid payable amount (Patient - Unknown Payable amount)

Result: The user's screen will be terminated, and the user's screen is not displayed.

7.2.2 Diagnostic Centres

Scenario 1:

Action: Searching for a valid diagnostic center (Patient)

Result: Diagnostic centers are known, and the Appointment screen is displayed.

Scenario 2:

Action: Searching for an invalid diagnostic center (Patient - Unknown

Diagnostic centers) Result: The Appointment screen will be terminated, and

the Appointment screen is not displayed.

Scenario 3:

Action: Entering a valid test (Patient)

Result: Test is known, and the Appointment

screen is displayed.

Scenario 4:

Action: Entering an invalid test (Patient - Unknown Test)

Result: The Appointment screen will be terminated, and the Appointment

screen is not displayed.

Scenario 5:

Action: Entering a valid doctor (Patient)

Result: Doctor is known, and the Appointment

screen is displayed.

Scenario 6:

Action: Entering an invalid doctor (Patient - Unknown Doctor)

Result: The Appointment screen will be terminated, and the Appointment screen is not displayed

8.CONCLUSION

The mobile application described is a comprehensive hospital management system designed to enhance healthcare services for patients, doctors, and administrators. Utilizing material design, the application boasts an attractive and intuitive interface with different screens catering to various modules, ensuring easy access to essential information and reports.

The application proves highly beneficial for patients seeking medical attention or undergoing treatments. One of its primary advantages is the elimination of waiting times by allowing patients to access their reports and receive necessary information even before their turn comes up at the clinic. Patients can access valuable health-related information, including precautions and measures to take for their well-being. This empowers patients to be more proactive in managing their health. Additionally, the user-friendly interface reduces the time and effort required to secure a doctor's appointment.

The application also caters to doctors, enabling them to manage patient records efficiently. They can access patient data and medical histories through the application, streamlining the diagnostic and treatment processes. With access to real-time patient information, doctors can provide better and more personalized care to patients, especially during emergencies when timely decision-making is crucial.

The application assists administrators in efficiently managing the hospital's operations. They can monitor patient flow, track appointments, and ensure smooth functioning of different departments. This streamlines administrative tasks, making the entire healthcare system more organized and effective.

Overall, the application strives to overcome the limitations of traditional hospital hospitalitiesystems. By as long as patients by appreciated statistics and personalized care, it enhances the overall patient experience. The focus on user-friendliness ensures that both patients and medical staff can use the application with ease. Additionally, the ability to cater to patient expectations allows the hospital to deliver a higher level of service, leading to increased patient satisfaction and loyalty. In conclusion, this hospital management mobile application serves as a cutting-edge tool that revolutionizes healthcare service delivery.

9.Future Enhancements

Indeed, leveraging enhanced technologies allows developers to create well-crafted applications with improved performance and enhanced features. Mobile application development frameworks provided developers with the tools to build applications on various platforms like iPhone, Android, Windows, and Blackberry, catering to a broader user base.

Ensuring security is a crucial aspect of application development. Implementing end-user authentication using APIs helps maintain the security of the application and mitigates the risk of hacking and unauthorized access. API-based authentication ensures that only authorized users can access sensitive data and features within the application.

To continuously improve the application's functionality, developers can focus on enhancing features based on user feedback and needs. Incorporating a robust chatting facility can foster better communication and engagement between users, leading to an improved user experience.

Additionally, by integrating health packages, the application can offer valuable health-related services, such as providing information about medicine and diseases. This enhances the application's utility and provides users with a comprehensive guide for managing their health. Utilizing advanced technologies and development frameworks allows developers to create secure, feature-rich, and userfriendly mobile applications. The integration of various functionalities, such as end-user authentication, improved chatting facilities, and health-related information, enhances the overall user experience and utility of the application. By continuously refining and enhancing the application, developers can ensure it meets the evolving needs of its users effectively.

Appendix-A

BIBLIOGRAPHY

WEB REFERENCES

<http://www.practo.com>

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Appendix-B

USER MANUAL

- The client gives all the data about the wellbeing offices so various modules will be consolidated as a different highlights for each module depiction.
- Client will give three decisions like trained professional, patient and executive.
- If client isn't selected then persevering ought to enroll in to the application and login by using a convenient number so an OTP will be given.
- After the OTP approval expect if we pick a patient, patient's dashboard will be shown.
- In course client will have an options of test game plans where the booked courses of action will appear, Fit assessment will give the google Fitbit of the patients, My Reports will give all the overview of reports, client can share the application and moreover can suggest a sidekick and even log out decision is consolidated.
- Patient's dashboard will manual for book the test by means of searching for the contaminations and picking the suggestive concentration by giving all the region, lab name, timings, layout of the lab, photos and subsequently book the test.
- Patients can in like manner book for another test, it gives the total aggregate so that sees whether they need home get and visiting date and time from home or the lab and a while later keep on paying.
- Patient's portion once-over will be shown by enrollment number, reference trained professional and the outright portion which are required and by picking the markdown code and the strategy for portion and trade put away which shows "Booked successfully".
- Food and wellbeing will demonstrate all the food things to be added by picking the food and sum, Exercises will guide us to speed like slow, medium and fast, and the distance and time to be referred to and even enter calories devoured and actually it is added.



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